MIMO Representations: Rosenbrock System Matrix and Matrix-fraction Descriptions

Multivariable Control, Homework 3

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The questions follow the lecture slides on representing multi-input multi-output systems:

- **State space from differential equations.** I converted a set of coupled differential equations into state-space form (slide 17).
- **System order.** I checked the proof that the order of an LTI system equals the degree of $|sI - A|$, i.e. its number of poles (slide 18).
- **Rosenbrock system matrix.** I found $P(s)$ for a polynomial description with zero initial conditions (slide 22) and discussed its advantages and drawbacks for multivariable systems.
- **A coprimeness condition.** I tested it through its second equivalent form (slide 51).
- **Matrix-fraction descriptions.** I computed the greatest common left divisor of an MFD (slide 61) and discussed the pros and cons of MFDs (slide 62).
- **Textbook exercises 2-3 and 2-8** (slide 63).